Adyapan School

Structural Biology

Duration - 2 months

Industry Certification

Skill India
N·S·D·C National Skill Development Corporation
Transforming the skill landscape

Skill India Certified

250+ Partner Companies

Adyapan School

From sequence to structure to function Master it all.

From molecular structures to biological insights - become industry-ready.

This immersive Structural Biology program takes you beyond foundational concepts into real-world biomolecular analysis. Learn protein structure determination, molecular modeling, visualization techniques, and structure-function relationships through hands-on tools and guided case studies. With a strong focus on practical workflows and research-driven applications, you'll graduate with the ability to analyze macromolecular structures, interpret biological mechanisms, and contribute confidently to advancements in biotechnology and drug discovery.

8	30+	20,000+
WEEKS	PROGRAMS OFFERED	STUDENTS

250+ PARTNERED COMPANIES

Meril

BioNome

ABOUT ADYAPAN SCHOOLS

Where education meets real-world impact

Not just a course — a platform to launch your career.

Adyapan Schools was built with a single conviction: learning works best when it happens in the real world. We partner with top companies, mentors, and industry platforms to ensure every student graduates with a portfolio of work that speaks louder than a certificate.

Our programs combine rigorous coursework with live client projects, giving you the skills and proof-of-work that employers actually want.

MISSION

To equip ambitious learners with practitioner-level digital marketing skills through mentor-led, project-based education that bridges the gap between learning and earning.

VISION

To be India's most trusted launchpad for the next generation of marketing leaders — defined not by degrees but by the real work.

Adyapan School

Everything you need to grow fast

PROGRAM HIGHLIGHTS

🚀 Live Industry Projects

Work on campaigns for real brands alongside your coursework. Build portfolio projects that prove your expertise to employers.

🎯 1-on-1 Mentorship

Dedicated mentors from Google, Microsoft, Mastercard and more. Get personalized guidance and industry connections.

📊 AI-Powered Marketing

Learn cutting-edge AI tools alongside evergreen fundamentals. Stay ahead of the curve in a rapidly evolving landscape.

🎓 Dual Certification

Earn both a Course Completion and Internship Certificate — accredited by Skill India Digital Hub and NSDC.

💼 Internship Guarantee

Graduate with an internship completion certificate from a live brand project. Concrete, resume-ready proof of work.

🌐 Industry Network

Join a network of alumni at Amazon, Google, Adobe, Microsoft. Access exclusive hiring events and referral opportunities.

CURRICULUM

8 weeks. 8 modules. Infinite impact.

WEEK 1

Introduction to Structural Biology

- Discussion of curriculum
- Understand the post-NGS era and why structural biology is critical in modern biological research
- Learn how to read and interpret biological macromolecules
- Explore the central sequence → structure → function relationship that governs molecular biology
- Establish proteins as the primary target study molecule
- Bridging the gap between gene sequences and three-dimensional biological function

WEEK 2

Protein Architecture

- Study the structure and functional roles of amino acids - the building blocks of all proteins
- Understand the four levels of protein structure
- Explore protein motifs and domains as recurring structural and functional units within diverse proteins
- Analyze conformational flexibility and how proteins adopt multiple structural states relevant to their function
- Understand the principles of protein folding - how a linear sequence spontaneously collapses into a stable three-dimensional structure

WEEK 3

Bonds & Energies in Macromolecules

- Understand covalent bonds in macromolecules
- Explore ionic and coordinate bonds and their contribution to structural stability and metal ion binding in proteins
- Learn how hydrophobic interactions drive protein folding by burying non-polar residues away from the aqueous environment
- Study Van der Waals forces - weak but collectively significant interactions that influence molecular packing and recognition
- Understand how the interplay of all these forces determines the final stable conformation and binding specificity of macromolecules

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WEEK 4

Enzymes: Structure & Function

- Introduction to enzymes - what makes them unique as biological catalysts
- Understand how enzyme active sites form "magic pockets" - shaped cavities that are precisely complementary to their substrates
- Explore enzyme–ligand interactions: binding modes, induced fit, lock-and-key models, and their structural basis
- Analyze the structure–function relationship in enzymes - how small changes in 3D structure can dramatically alter catalytic activity
- Study how structural knowledge of enzymes is applied in drug discovery, designing inhibitors that fit and block active sites

WEEK 5

Structural Biology Techniques

- Get an overview of the three major techniques for determining macromolecular structures
- Understand the basic principles behind macromolecular crystallography - how crystals of proteins are formed and used to obtain diffraction data
- Learn the fundamentals of Nuclear Magnetic Resonance (NMR) - how it probes atomic-level structure and dynamics of molecules in solution
- Explore Cryo-Electron Microscopy - how flash-frozen samples are imaged at near-atomic resolution without the need for crystals
- Conduct a comparative discussion of all three methods

WEEK 6

Macromolecular Crystallography: Crystallization & Data Collection

- Understand the principles of protein crystallization - what conditions drive ordered crystal lattice formation from solution
- Explore methods of protein crystallization: hanging drop, sitting drop, microbatch, and counter-diffusion techniques
- Study the thermodynamics and kinetics of protein crystallization - nucleation, crystal growth, and how to optimize conditions
- Learn how X-ray diffraction data is collected from protein crystals using synchrotron sources and laboratory diffractometers

CURRICULUM

8 weeks. 8 modules. Infinite impact.

WEEK 7

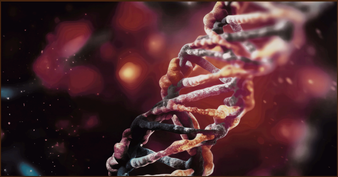
Macromolecular Crystallography: Data Analysis & Structure Determination

- Understand X-ray diffraction patterns and apply Bragg's equation
- Atomic scattering factors and structure factor expressions - the mathematical framework linking electron density to diffraction data
- Explore the phase problem - why phases cannot be directly measured and why solving it is the central challenge of crystallography
- Study methods for phase determination
- Understand the complete pipeline of macromolecular structure determination

WEEK 8

3D Structure Visualization & Introduction to Molecular Dynamics

- Understand clinical genomics applications: rare disease diagnosis, pharmacogenomics, and cancer genomics
- Study genome-wide association studies (GWAS)
- Explore polygenic risk scores (PRS) and their application in personalised medicine and disease risk prediction
- Introduction to CRISPR-based genomic editing and its intersection with functional genomics
- Understand ethical, legal, and social issues (ELSI) in genomics



Adyapan School

WHO THIS IS FOR

This course is perfect for

- Students & Career Switchers (Biology, Biotechnology & Life Sciences)
- Aspiring Structural Biologists & Molecular Researchers
- Biotechnology, Biochemistry & Molecular Biology Students
- Researchers Working in Protein Structure & Function Analysis
- Bioinformatics & Computational Biology Enthusiasts
- Drug Discovery & Pharmaceutical Research Aspirants

CERTIFICATIONS

ALUMNI NETWORK

Our alumni work at world-class companies

Amazon | Adobe | Google | Autodesk | Microsoft | Deloitte

Your career switch is one click away.

Ready to begin? Apply at adyapanschool.com or email us at support@adyapan.com

Apply Now